Supplementary Information

Supplementary information for "Evolutionary dynamics of nascent multicellular lineages" (2024), Doulcier G, Remigi P, Rexin D and Rainey PB.

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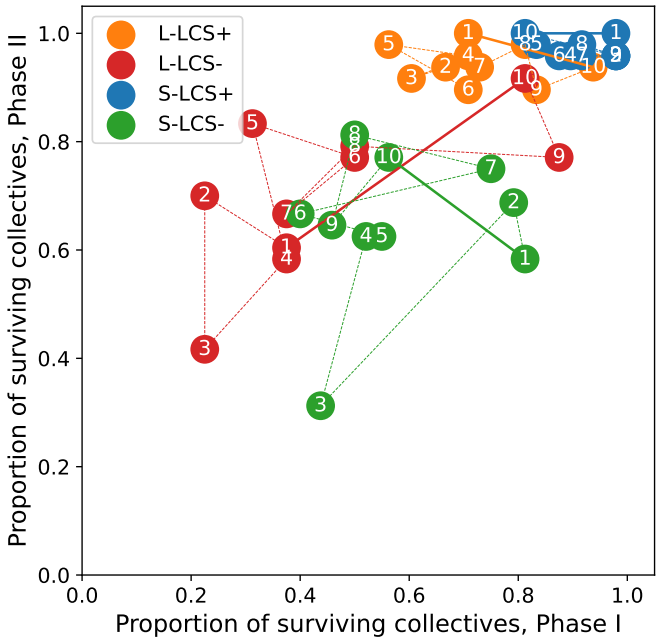


Figure S1. Proportion of surviving lineages in Ph I and Ph II for each cycle. Each circle corresponds to the proportion of surviving lineages for the cycle that is labelled. Each color is a different treatment, LCS$^+$: Lineage 17 from (1) with a life cycle switch, LCS$^-$: same lineage with switching capacity eliminated, S: standard microcosms, L: large microcosms. The dashed lines link consecutive cycles, the thick lines link the first and last cycle, representing the resulting change over the whole experiment. Note that for all treatment but L-LCS$^-$, there exists a trade-off between survival in the two phases.

Supplementary Note 1: Fitness assays

Previously, measures of fitness have involved lineage-level competition experiments against ancestral types that are often difficult to interpret and challenging to conduct. Observation of the genealogies of lineages as shown in Figure 3 led to realisation that the probability of avoiding extinction is a good predictor of future evolutionary success. Henceforth, measures of fitness were determined, not by competition with an ancestral type, but by measuring the number of lineage extinction events after replicate lineages of single test genotypes have gone through a further life-cycle generation.

New experiments were conducted with the same protocol as for the main lineage-selection experiment and performed for a single cycle, each time starting with a population of eight microcosms founded by the same genotype. At the end of phase I, the presence of a mat was recorded and the density of SM cells was measured though plating. At the end of phase II, the density of WS cells was estimated through plating.

This experiment was conducted on average 44 times for each of the 48 genotype and microcosm size combination tested (hence a total of 2120 microcosms). For each genotype, we estimated the probability of three events: M the production a mat at the end of phase I, S_I the production of cells with the switched phenotype SM cells at the end of Ph I, S_{II} the production cells with the switched phenotype WS at the end of Ph II. Further, we defined the "fitness" of a genotype as its probability to survive a full two-phase cycle, as the probability to survive both phases, hence $F = \mathbf{P}(M \wedge S_I \wedge S_{II})$.

We estimated the three probabilities by maximum of likelihood on a beta-binomial model, with a non-informative prior (uniform in [0,1]). As a consequence, ff there were n independent trails and k successes, the estimate of probability is simply the proportion of successes $\hat{p} = k/n$, and the Bayesian confidence interval is given by the quantiles of a beta distribution parametrised as Beta$(1+k, 1+n-k)$.

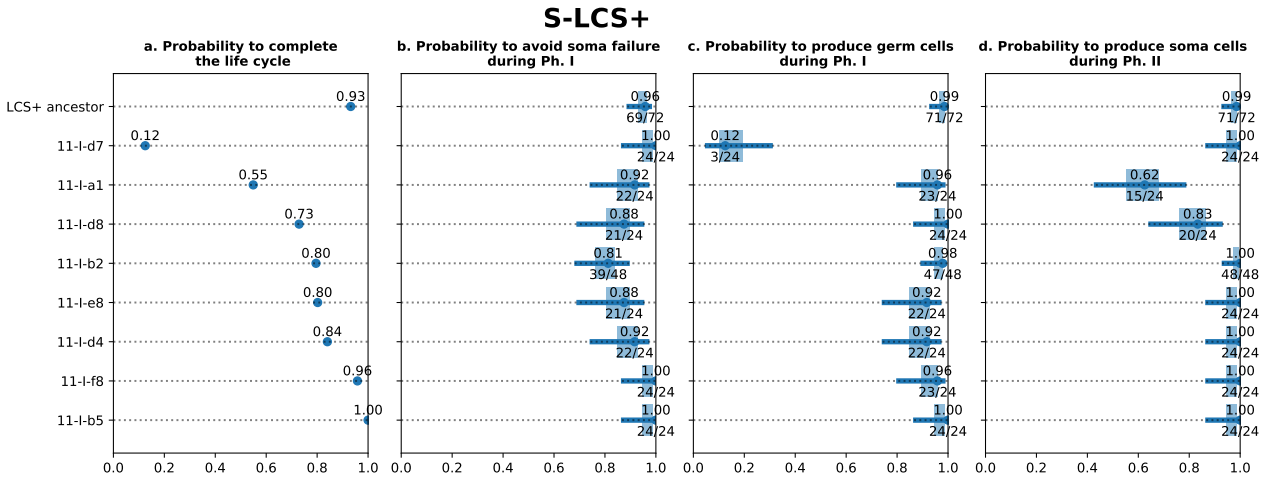


Figure S2. Probabilities in S-LCS$^+$ endpoints. Each line corresponds to a genotype derived from the ancestor LCS$^+$, and grown in small microcosms. Same caption than Figure 7

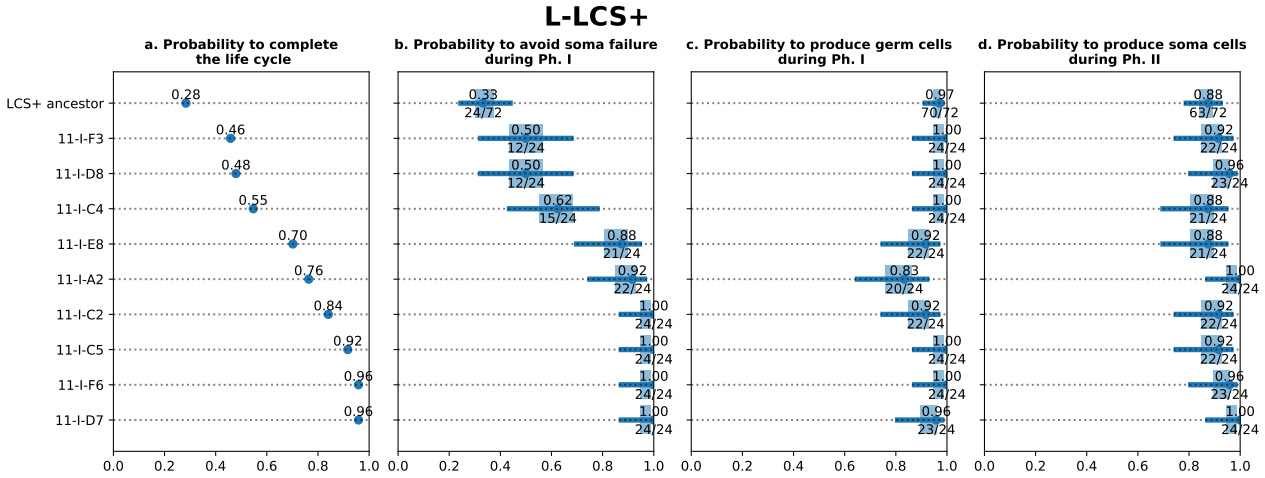


Figure S3. Probabilities in L-LCS$^+$ endpoints. Each line corresponds to a genotype derived from the ancestor LCS$^+$, and grown in large microcosms. Same caption than Figure 7

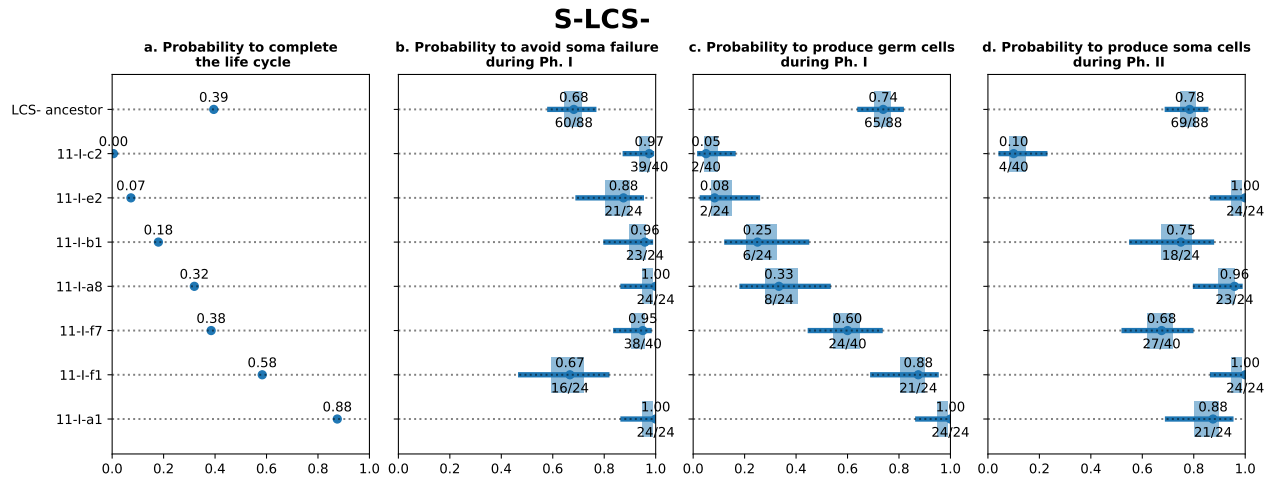


Figure S4. Probabilities in S-LCS$^-$ endpoints. Each line corresponds to a genotype derived from the ancestor LCS$^-$, and grown in small microcosms. Same caption than Figure 7

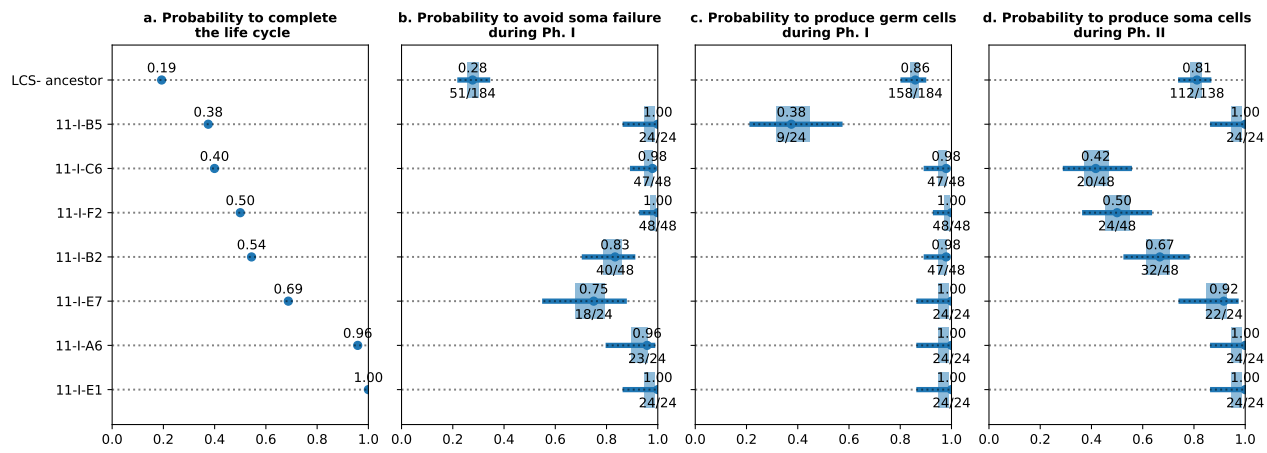


Figure S5. Probabilities in L-LCS- endpoints. Each line corresponds to a genotype derived from the ancestor LCS$^-$, and grown in large microcosms. Same caption than Figure 7

References.

[1] Hammerschmidt K, Rose CJ, Kerr B, Rainey PB. Life cycles, fitness decoupling and the evolution of multicellularity. Nature. 2014 Nov;515(7525):75-9. Available from: http://www.nature.com/nature/journal/v515/n7525/abs/nature13884.html.

[2] Efron B. Bayes' Theorem in the 21st Century. Science. 2013 Jun;340(6137):1177-8. Available from: http://science.sciencemag.org/content/340/6137/1177.

[3] Koller D, Friedman N. Probabilistic graphical models: principles and techniques. MIT press; 2009.

[4] Dempster AP, Laird NM, Rubin DB. Maximum Likelihood from Incomplete Data via the EM Algorithm. Journal of the Royal Statistical Society Series B (Methodological). 1977;39(1):1-38. Available from: http://www.jstor.org/stable/2984875.

Supplementary Note 2: Adaptive value estimation

We describe a method to assign to each collective within the genealogy a most probable adaptive value. In technical terms, the "most probable adaptive value" refers to a maximum likelihood estimation (MLE) that is a way to assign, under a defined graphical probabilistic model, a value to a parameter that maximise the probability of the observed result. The adaptive value is here a hidden variable: one does not simply observe the adaptive value of a collective, but performs experiments that provide information on its true value.

This endeavour requires first encoding of both the observations and the hidden characteristics of the collectives in a model, that takes the form of a joint probability distribution. Second, a method to find the most likely assignation of an adaptive value to each culture within the genealogy, for a given mutation rate, will be presented. Conversely, parameter inference allows assignment of the most likely value of a mutation rate for a given configuration. Finally, the iterative method of Expectation-Maximisation will be used to concurrently fit to the data both the adaptive value and the mutation rate.

A. Model structure.

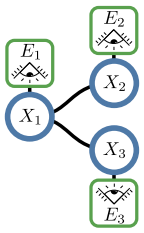


Figure S6. Bayesian Network. Bayesian network corresponding to a genealogy of three collectives. Collective 1 has two offsprings: collectives 2 and 3. Vertices are random variables, edges are conditional dependence. The hidden variables X encode the survival probability of the collectives, while the observed variables E encode the outcome of the experiment (survived or not).

Each collective genealogy $\mathcal{T} = (V, \mathcal{E})$ is a forest containing N numbered vertices: $V = 1...N$ which are the collectives, and an edge linking each collective i to its parent $p_i : \mathcal{E} = (i, p_i)_{i=1...N}$.

The model assigns to each collective an observed and a hidden variable. For a collective i, let E_i be the random variable encoding whether this collective went extinct ($E_i = 1$) or not ($E_i = 0$). In the field of probabilistic graphical models the observed values such as the $(E_i)_{i=1...N}$ are called "evidence".

The first assumption of the model is that there exists a hidden random variable for each collective i, called X_i, representing its survival probability. In probabilistic terms, this means that the conditional probability of extinction of a collective i knowing the value of the survival probability x_i is exactly the value of x_i:

$$\mathbb{P}(E_i = e_i | X_i = x_i) = \begin{cases} x_i & \text{if } e_i = 0 \\ 1 - x_i & \text{if } e_i = 1 \end{cases} \tag{1}$$

The second assumption of the model is that the survival probability of a collective is transmitted from one collective generation to the next, with an unbiased small variation. In probabilistic terms it means that if the parent collective has survival probability x_{p_i}, then the survival probability of its offspring collective follows a distribution with mean x_{p_i} Several probability distributions families can be used. Colgen implement two of them: a beta distribution and a "jump" distribution family parameterized by a single scalar σ.

For the beta distribution:

$$\mathbb{P}(X_i \in A | X_{p_i} = x_{p_i}) = C \int_A \frac{x^{\sigma x_{p_i}} (1-x)^{\sigma(1-x_{p_i})}}{B(1+\sigma x_{p_i}, 1+\sigma(1-x_{p_i}))} \tag{2}$$

with $A \subset [0,1]$, $B(u,v) = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}$, Γ the Gamma function, and C a scaling factor ensuring that $\int_0^1 \mathbb{P}(X_i = x | X_{p_i} = x_{p_i}) dx = 1$.

For the jump distribution:

$$\mathbb{P}(X_i = p_i | X_{p_i} = x_{p_i}) = \begin{cases} \sigma + (1-\sigma)C & \text{if } X_i = x_{p_i} \\ (1-\sigma)C & \text{otherwise.} \end{cases} \tag{3}$$

With x_i, x_{p_i} both in $(1/C, 2/C ... k/C, 1)$, and C a scaling factor ensuring that $\int_0^1 \mathbb{P}(X_i = x | X_{p_i} = x_{p_i}) dx = 1$.

During the experiment, the values of $E_i = e_i$ (extinct or not) are observed, whereas the values of X_i are "hidden".

B. Bayesian inference.

Consider a collective i with associated survival probability X_i and experimental outcome E_i. The prior probability distribution of X_i is given by Equation 2. Consider for an instant that x_{p_i}, the adaptive value of its parent is a fixed parameter. Bayes' Theorem gives the expression of the probability of X_i, conditional on the observed value of E_i:

$$\mathbb{P}(X_i|E_i) = \frac{\mathbb{P}(E_i|X_i)\mathbb{P}(X_i)}{\mathbb{P}(E_i)} \tag{4}$$

Where $\mathbb{P}(X_i)$ is the prior probability of X_i, $\mathbb{P}(E_i|X_i)$ is the observation model from Equation 1 and $\mathbb{P}(E_i)$ is the probability of the evidence. The maximum likelihood estimate (or more precisely the maximum a posteriori estimate, MAP) of the value of X_i is given by taking the value of x that maximises $\mathbb{P}(X_i = x|E_i)$, note that the value of the MAP does not depend on $\mathbb{P}(E_i)$, which is a simple scaling constant. Observing the outcome of the experiment (survival or extinction) has an effect on the posterior: the whole distribution (and consequently the MAP) is shifted toward higher values if the collective survived, or lower values if the collective went extinct.

The problem lies, as often in Bayesian statistics, in the prior probability of X_i (2). Indeed, in the context of the experiment, this probability depends on the state of all the ancestors of the collective i. Applying Bayes theorem to all those interdependent observations quickly becomes a complex undertaking without a proper method. Evidence propagation, presented hereafter, is this optimal method.

The next section details the technical steps required to find the most probable values of X_i throughout the genealogy.

C. Finding the most likely configuration of the genealogy.

The configuration of the full genealogy is the knowledge of the state of all vertices. Hence, it is encoded as a joint distribution that maps each configuration to a probability:

$$(\mathbf{x}, \mathbf{e}) \mapsto \mathbb{P}(\mathbf{X} = \mathbf{x}, \mathbf{E} = \mathbf{e}) \tag{5}$$

Where bold letters are used to represent vectors of dimension N, $\mathbf{x} = (x_i)_{i\in V}$, $\mathbf{X} = (X_i)_{i\in V}$, $\mathbf{E} = (E_i)_{i\in V}$, $\mathbf{e} = (e_i)_{i\in V}$.

The likelihood of a given set of adaptive values $\mathbf{x}$ is defined as the probability of observing this value, conditional on the state of all collectives:

$$L(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x}|\mathbf{E} = \mathbf{e}) \tag{6}$$

The maximum a posteriori configuration of the network is given by finding the value of $\mathbf{x}$ that maximises $L(\mathbf{x})$. The naïve, brute force method to find this value is to compute the L for all $\text{Card}(\omega)^N$ possible configurations (where $\text{Card}(\omega)$ is the number of possible value of X. If the adaptive value is discretised in a hundred of values, this gives 100^{48} computations (in the order of the number of atom of earth $\approx 10^{50}$, and way too much even for modern computers).

However, the probability distribution can be efficiently factorised thanks to the definition of conditional probabilities whose structure is represented in the graph: each pair of nodes not linked by an edge is conditionally independent of one another. As an example consider the small network in Figure S6. The factorisation will go like this:

$$\begin{aligned}
\mathbf{x}^* &= \arg\max_{\mathbf{X}} \mathbb{P}(X_1, X_2, X_3|E_1, E_2, E_3) \\
&= \arg\max_{\mathbf{X}} \frac{\mathbb{P}(X_1, X_2, X_3, E_1, E_2, E_3)}{\mathbb{P}(E_1, E_2, E_3)} \quad \text{(Definition of conditional probability)} \\
&= \arg\max_{\mathbf{X}} \mathbb{P}(X_1, X_2, X_3, E_1, E_2, E_3) \quad \text{(Denominator constant with respect to } \mathbf{x}) \\
&= \arg\max_{\mathbf{X}} \mathbb{P}(X_1)\mathbb{P}(E_1|X_1)\mathbb{P}(X_2|X_1)\mathbb{P}(E_2|X_2)\mathbb{P}(X_3|X_1)\mathbb{P}(E_3|X_3) \quad \text{(Conditional independence)} \\
&= \arg\max_{X_1} \left[\underbrace{\mathbb{P}(X_1)\mathbb{P}(E_1|X_1)}_{\phi_1(X_1)} \underbrace{\arg\max_{X_2} (\mathbb{P}(X_2|X_1)\mathbb{P}(E_2|X_2))}_{\phi_2(X_1)} \underbrace{\arg\max_{X_3} (\mathbb{P}(X_3|X_1)\mathbb{P}(E_3|X_3))}_{\phi_3(X_1)} \right] \\
&= \arg\max_{X_1} \phi_1(X_1)\phi_2(X_1)\phi_3(X_1) \quad \text{(Variable elimination of } X_1, X_2)
\end{aligned}$$

Additionally, the "maximum" operation is distributive, which allows for efficient computation of local maxima (ϕ_2, ϕ_3, called messages) that depend on a single variable. This constitutes the essence of the Max-Product Belief propagation algorithm ((3), Chapter 13).

D. Expectation-Maximisation procedure.

Now, how can one find the value of the parameters of the model ? The Expectation-Maximisation (EM) algorithm (4), which might be one of the most influential advances in modern computational statistics and is widely used in Gaussian Mixtures, and Hidden Markov Models, provides an iterative solution.

First, note that for any given configuration of the network (values of $\mathbf{X}$ and $\mathbf{E}$), finding the maximum likelihood estimate of the parameter (mutation rate σ) is a simple matter of computing the value of the parameter that maximises the likelihood $\mathbb{P}(\mathbf{X}, \mathbf{E})$).

Now, starting from an arbitrary value of parameters (say, $\sigma = \frac{1}{2}$), one may compute the Maximum A Posteriori configuration of the network, that assigns a value to each hidden variable. This constitutes the Expectation (E) step. Then, a new value of the parameter is computed using the Maximum of Likelihood estimate. This constitutes the maximisation step (M). The procedure can be iterated until convergence. This algorithm is guaranteed to reach a local maximum in likelihood, but may not reach the global maximum. However, its conceptual simplicity means that it is often the best heuristic available for practical applications.

E. Pre-processing specific to the present life-cycle experiment.

The mat-selection experiment described in the main text features two alternating phases (selection for germ, selection for soma) involving very different processes and are likely to yield different survival probabilities. As a consequence, two sub-trees were extracted for each genealogy, one for Ph I and the other for Ph II (Figure S7) and analysed separately.

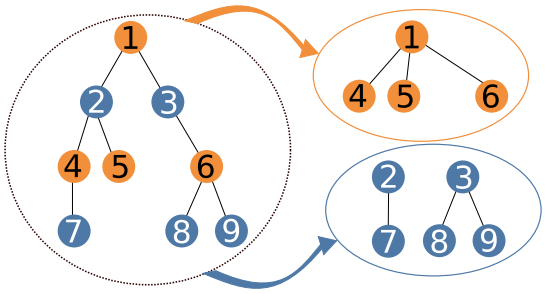


Figure S7. Subtree extraction procedure. Phase 1 (selection for a somatic mat and germ-like cells) and phase 2 (selection for the reversion to soma-like cells) are qualitatively different and are treated separately. In order to achieve this, each collective genealogy is split into a phase 1 and a phase 2 tree that are analysed on their own, and re-assembled for visualisation.

References.

[1] Hammerschmidt K, Rose CJ, Kerr B, Rainey PB. Life cycles, fitness decoupling and the evolution of multicellularity. Nature. 2014 Nov;515(7525):75-9. Available from: http://www.nature.com/nature/journal/v515/n7525/abs/nature13884.html.

[2] Efron B. Bayes' Theorem in the 21st Century. Science. 2013 Jun;340(6137):1177-8. Available from: http://science.sciencemag.org/content/340/6137/1177.

[3] Koller D, Friedman N. Probabilistic graphical models: principles and techniques. MIT press; 2009.

[4] Dempster AP, Laird NM, Rubin DB. Maximum Likelihood from Incomplete Data via the EM Algorithm. Journal of the Royal Statistical Society Series B (Methodological). 1977;39(1):1-38. Available from: http://www.jstor.org/stable/2984875.

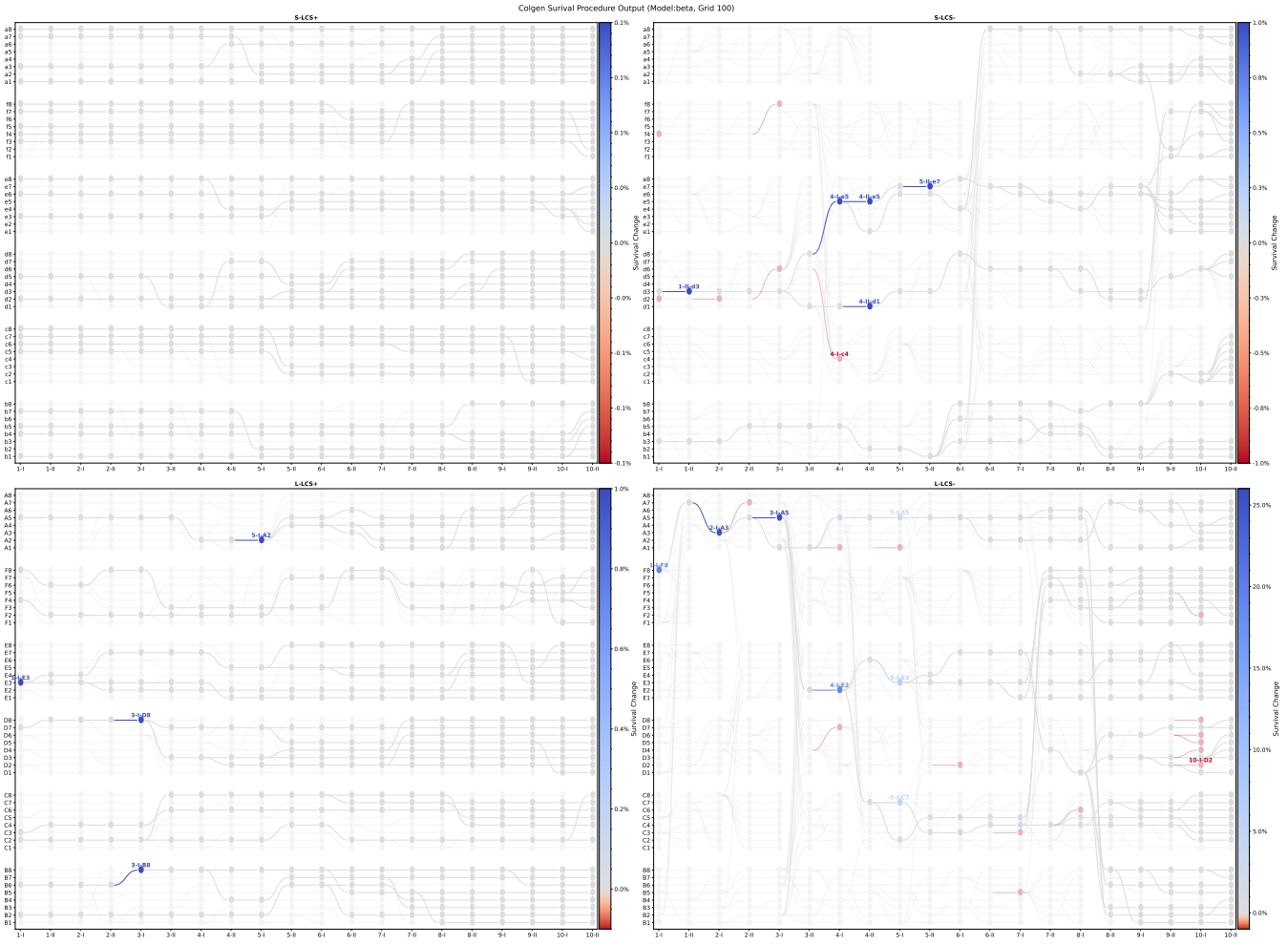


Figure S8. Nodes in the genealogies predicted to mark adaptive change. Model Beta, Grid:100, Sigma-max:1500.

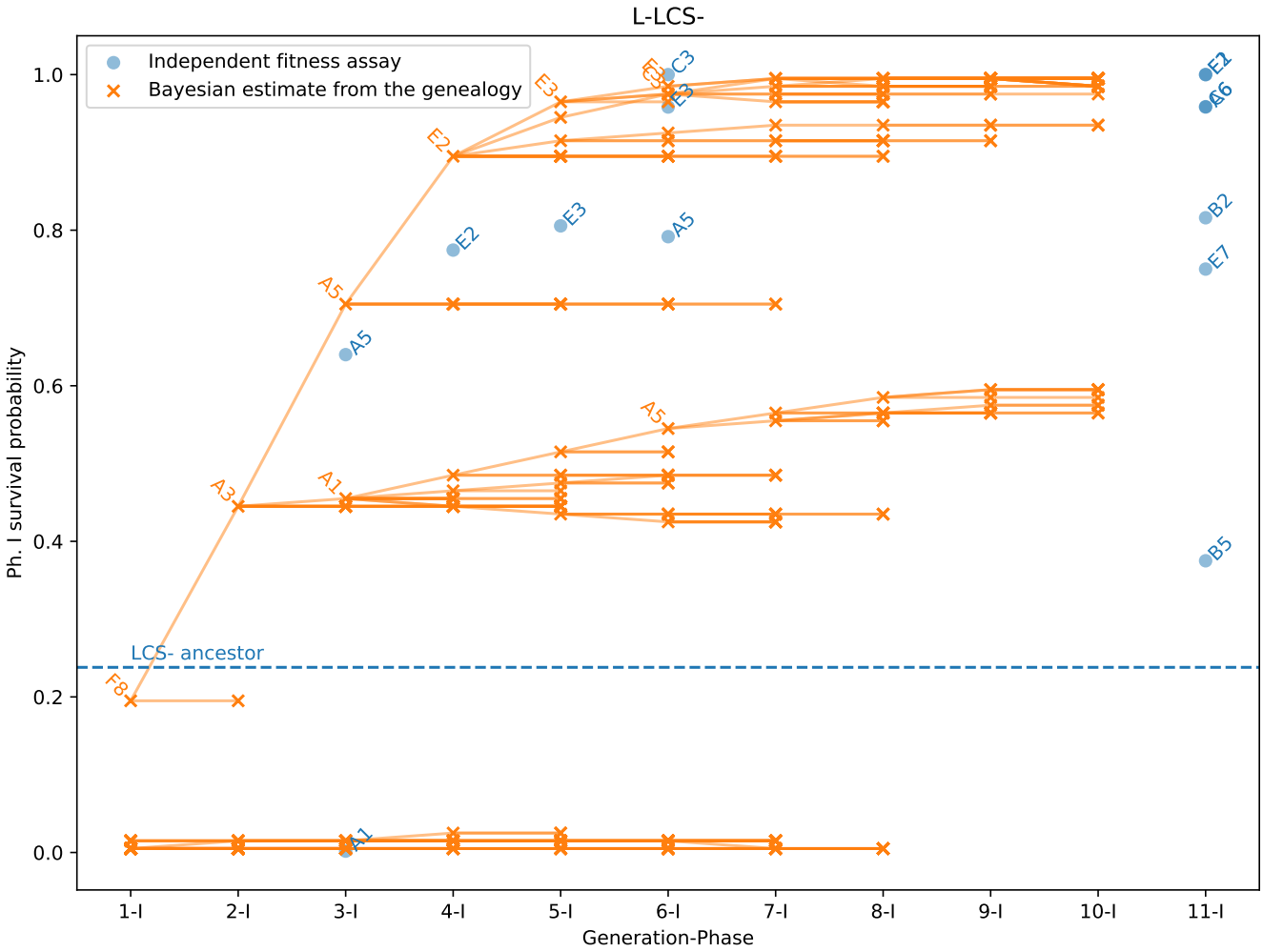


Figure S9. Phase I survival probability as estimated by independent fitness assays (blue, data from Figures 7 and S5) and the Bayesian beta model of Colgen (Orange, Sigma=1500, Grid=100, from Figure 4).

Supplementary Note 3: Sequencing and mutation history reconstruction

A. Number of mutation identified by direct sequencing.

Figure S10 represents the number of mutations identified in each sequenced lineage. Note that the L-LCS$^-$ has a significantly higher number of mutations than S-LCS$^-$ at cycle 10 depsite being started with the same genotype. It seems that it has became a hypermutator line.

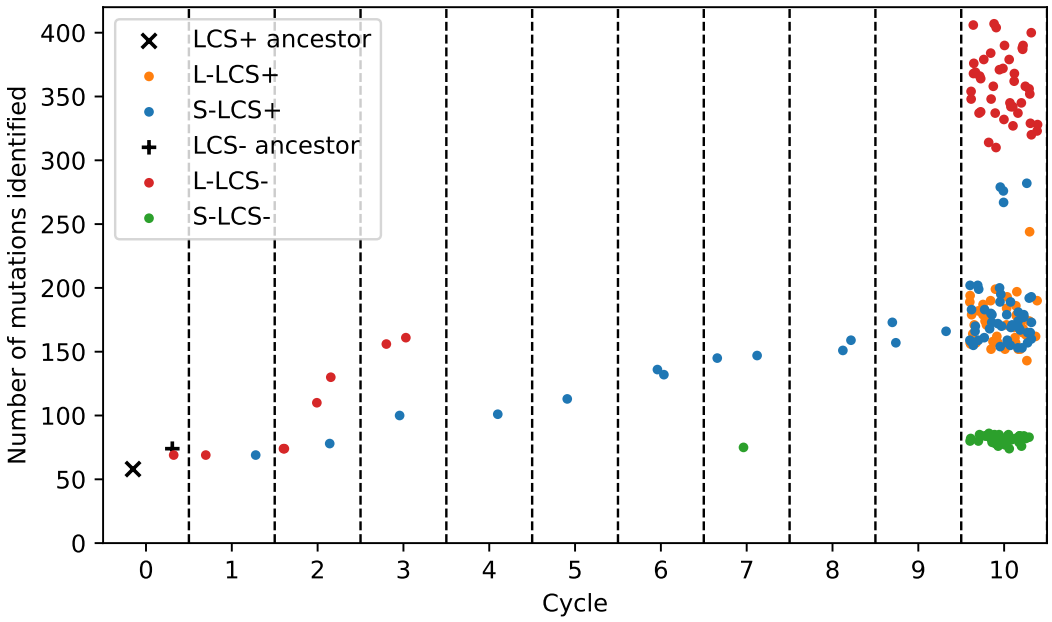


Figure S10. Number of mutations identified by direct sequencing. Each circle is a sequenced culture. Mutations are identified by mapping shorts read onto the reference SBW25 genome (GCF000009225_2). See materials and methods for details. The x-axis correspond to the cycle, the x-position is randomized within each cycle to avoid overlap. Compare this figure with Figure 5 (Note that in Figure 5, the mutations present in the ancestor are filtered out) and Figure S11 that show this data propagated through the whole genealogy.

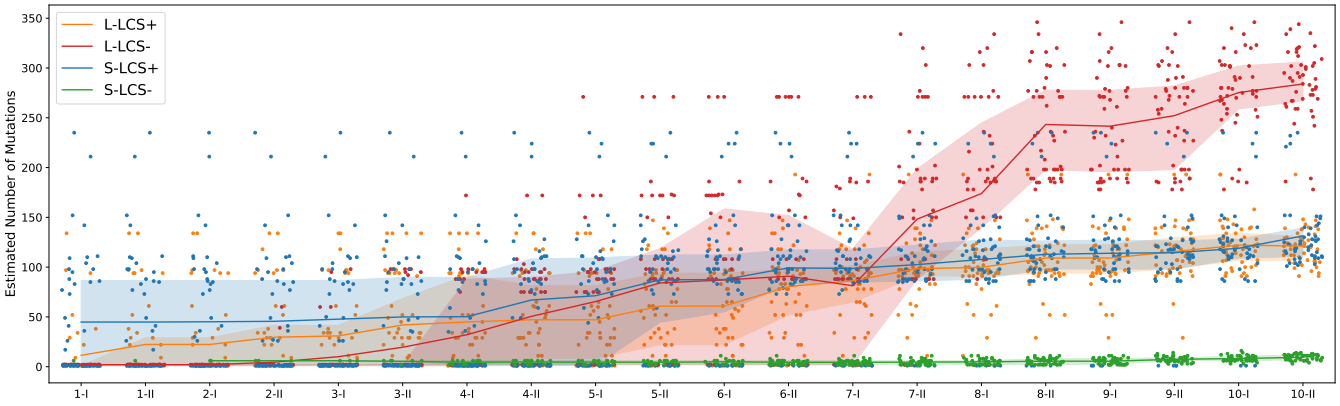


Figure S11. Number of new mutations reconstructed by belief propagation. Each circle is a sequenced WS genotype. The x-axis corresponds to generation of the life cycle, the x-position is randomized within each cycle to avoid overlap. Compare this figure with Figure S10 that shows these data only for mutations identified by sequencing (and includes mutations present in the ancestor). Lines correspond to the average number of mutations per treatment, shaded areas correspond to 50%-quantils.

B. Propagation of mutational data from sequencing through genealogies.

We use the belief propagation algorithm presented in Supplementary note 2, but the hidden variable X_i is now the presence ($X_i = 1$) or absence ($X_i = 0$) of the mutation within the lineage. The observed variable E_i is replaced by a conditionally Bernoulli variable O_i encoding the event "the mutation of interest was identified in sequences obtained from collective i" for the genotype that was sequenced. The observation model must be specified:

$$\mathbb{P}(0_i = i | X_i = j) = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}_{i,j} \quad (7)$$

Where p and q represent, respectively, the probability of true negative and true positive reporting. In the following $p = q = 1$. Finally, the heredity model is specified as:

$$\mathbb{P}(X_i = i | X_{p_i} = j) = \begin{bmatrix} 1-\mu & \mu \\ \mu_r & 1-\mu_r \end{bmatrix}_{i,j} \quad (8)$$

Where μ is the probability that the mutation appears within one cycle and μ_r the probability of losing the mutation. In the main text, arbitrary values of $\mu = 0.001$ and $\mu_r = 0.00001$ have been selected.

C. Number of mutations.

Figures S11 and S13 represent the number of mutations predicted in each node by the belief propagation algorithm.

D. Mut mutations.

We focused on the mutations in genes known to encode components of the methyl-directed mismatch repair system (*mut* family: *mutS*, *mutL*, *mutY* and *mutM*, 17 mutations were identified, see Table 1). The results are shown on the genealogies in Figures S14 to S17.

Table 1. Description of the mutations in the methyl-directed mismatch repair system identified in the dataset. Figure S12 shows in which lineages they were identified.

Gene	Position	Mutation	Annotation	Description
mutY ←	355688	C→T	G153D (GGC→GAC)	A/G-specific adenine glycosylase
mutL →	586966	C→T	S5L (TCG→TTG)	DNA mismatch repair protein
mutL →	587050	A→G	E33G (GAG→GGG)	DNA mismatch repair protein
mutL →	587260	T→G	L103R (CTG→CGG)	DNA mismatch repair protein
mutL →	587555	C→T	L201L (CTC→CTT)	DNA mismatch repair protein
mutL →	587737	A→G	N262S (AAC→AGC)	DNA mismatch repair protein
mutL →	588126	(G)5→6	coding (1174/1902 nt)	DNA mismatch repair protein
mutL →	588551	C→T	A533A (GCC→GCT)	DNA mismatch repair protein
mutS ←	1300356	C→T	G707S (GGC→AGC)	DNA mismatch repair protein
mutS ←	1300673	+A	coding (1802/2592 nt)	DNA mismatch repair protein
mutS ←	1300783	A→G	R564R (CGT→CGC)	DNA mismatch repair protein
mutS ←	1300986	T→G	T497P (ACC→CCC)	DNA mismatch repair protein
mutS ←	1301392	A→G	R361R (CGT→CGC)	DNA mismatch repair protein
mutS ←	1302049	A→G	R142R (CGT→CGC)	DNA mismatch repair protein
mutS ←	1302080	T→C	N132S (AAC→AGC)	DNA mismatch repair protein
mutS ← / ← PFLU_1165	1302576	G→A	intergenic (-102/+68)	DNA mismatch repair protein/putative membrane protein
mutM ←	6343361	A→G	S46P (TCC→CCC)	formamidopyrimidine-DNA glycosylase

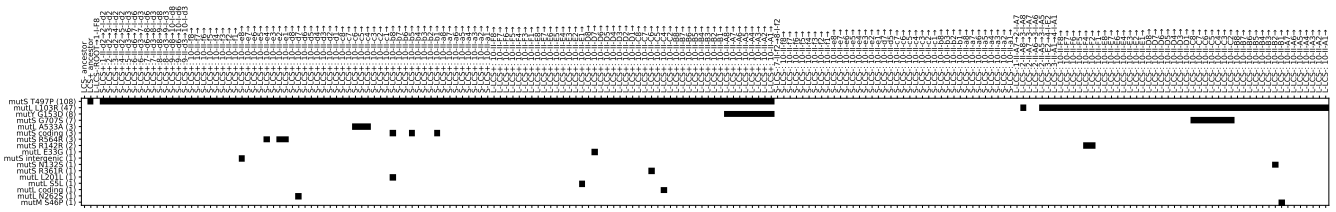


Figure S12. Identified mutations of the methyl-directed mismatch repair system in the sequenced genomes. A black square indicate that the mutation was identified. See Table 1 for a description of the mutations.

E. Finding the location of the L103R *mutL* mutation.

Figures S18 to S20 explain how we pinpoint the location of L103R *mutL*.

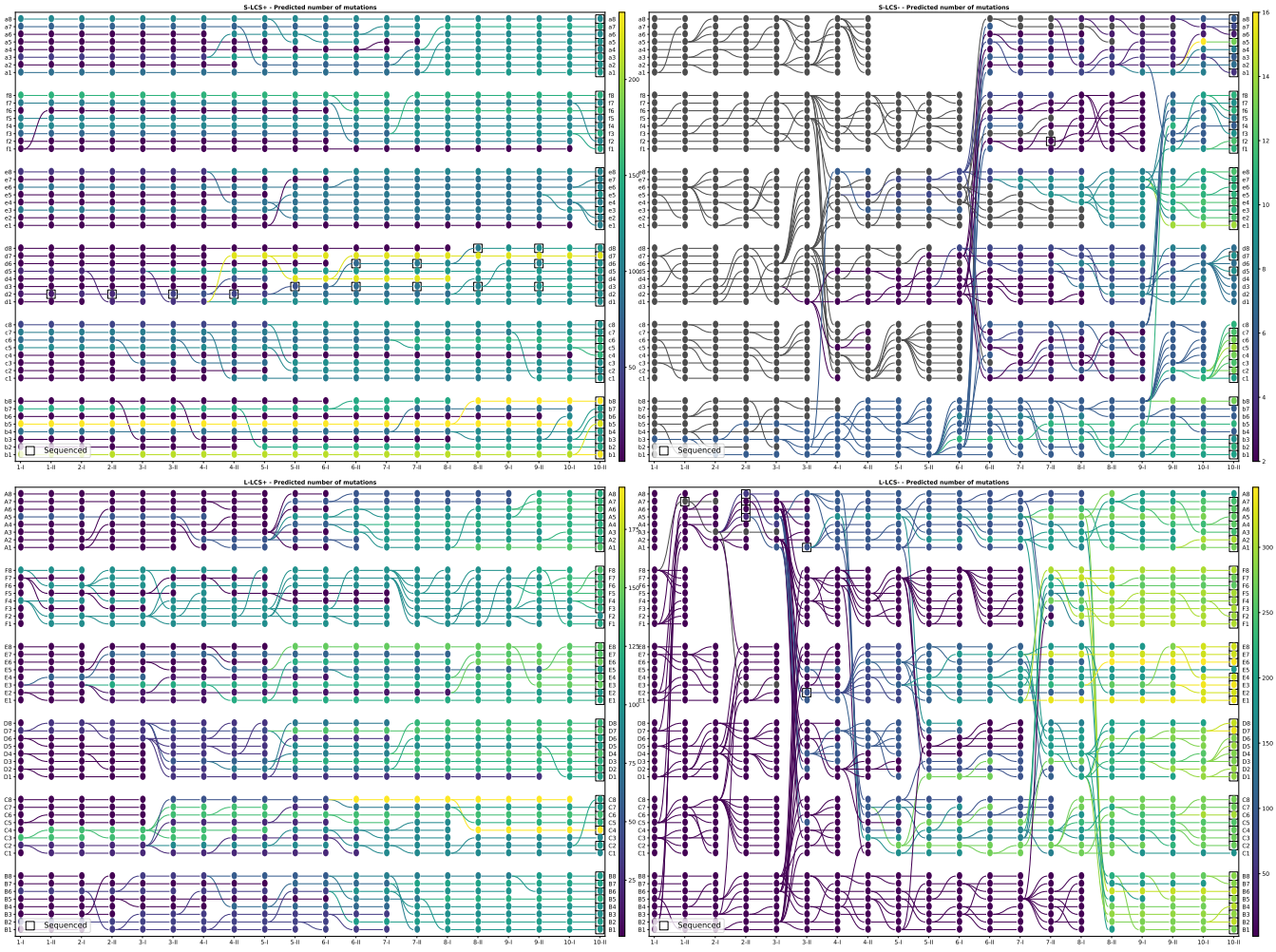


Figure S13. Estimated number of mutations per node. Sequenced genotypes are marked with a black square.

When we only had the endpoints of L-LCS$^-$ sequenced, the L103R *mutL* mutation was in all the sequenced nodes, so the best prediction that could be done was "in the last common ancestor" See Figure S18.

However, the information obtained through the survival estimation algorithm prompted further investigation around 2-II-A5 and 3-II-E2, so additional sequencing was conducted of nodes 1-II-A7, 2-II-A5, 2-II-A6, 2-II-A7, 2-II-A8, 3-II-E2 and 3-II-A1. The mutation is present in 2-II-A5, 2-II-A8, 3-II-E2 and 3-II-A1 and absent in the other nodes (Figure S19).

The propagation procedure of `colgen` output two origins of mutL (Figure S20), in 2-II-A5 and 2-II-A8, because it is highly unlikely that it was in 1-I-A3 and was not transmitted to 2-II-A6 and 2-II-A7.

This mean that the mutation probably appeared between the single cell bottleneck at the end of 1-II-A7 and the single cell bottleneck at 2-II-A5 and 2-II-A8. However, not all cells within 1-I-A3 carry the mutation.

F. Mutation bias in *mutY*.

Figure S21 shows the mutation biais in lineages with a *mutY* mutation.

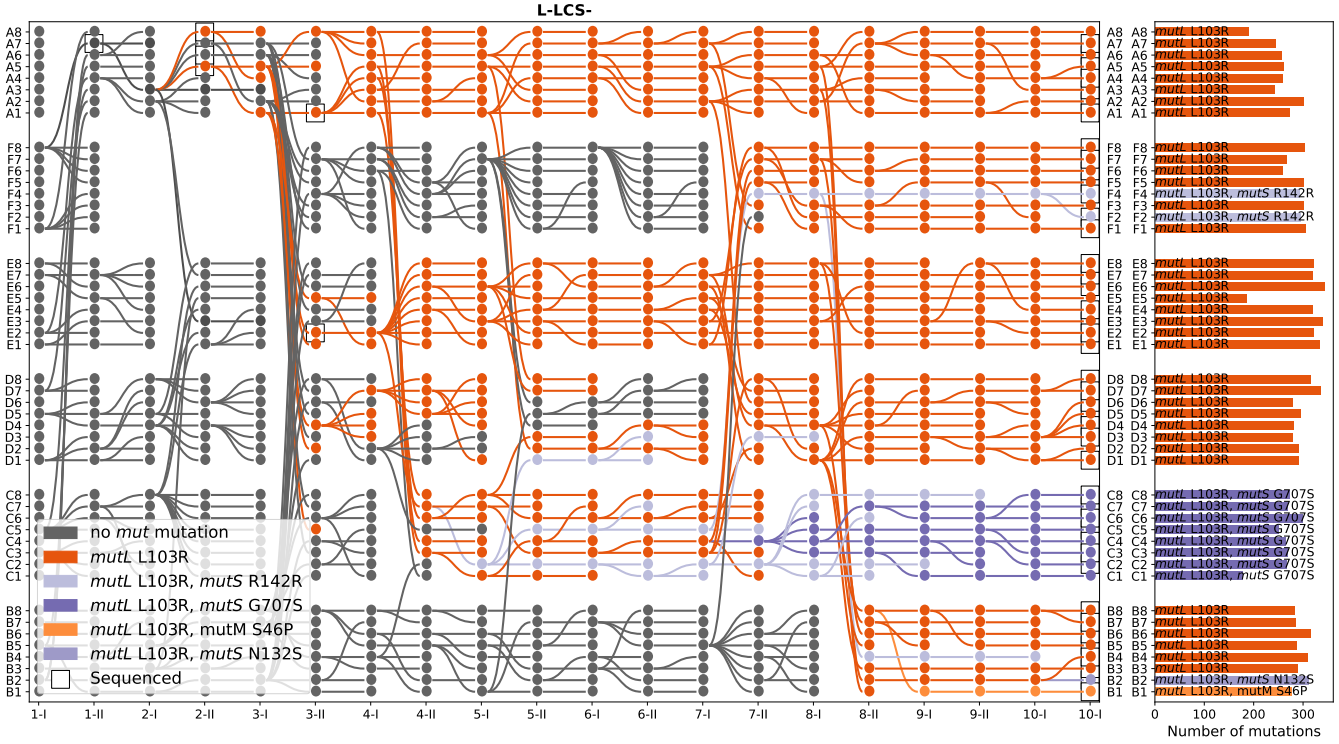


Figure S14. Mut mutations and number of mutations at the endpoint for L-LCS$^-$

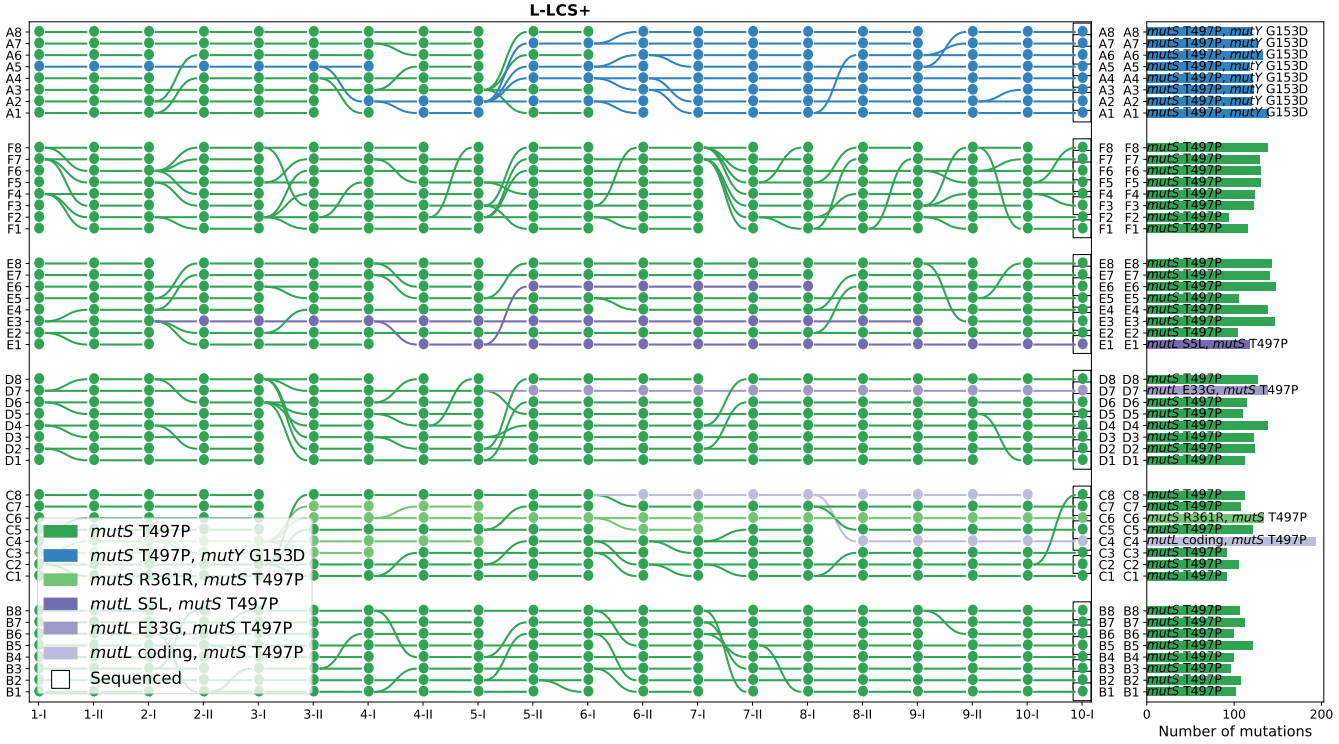


Figure S15. Mut mutations and number of mutations at the endpoint for L-LCS$^+$

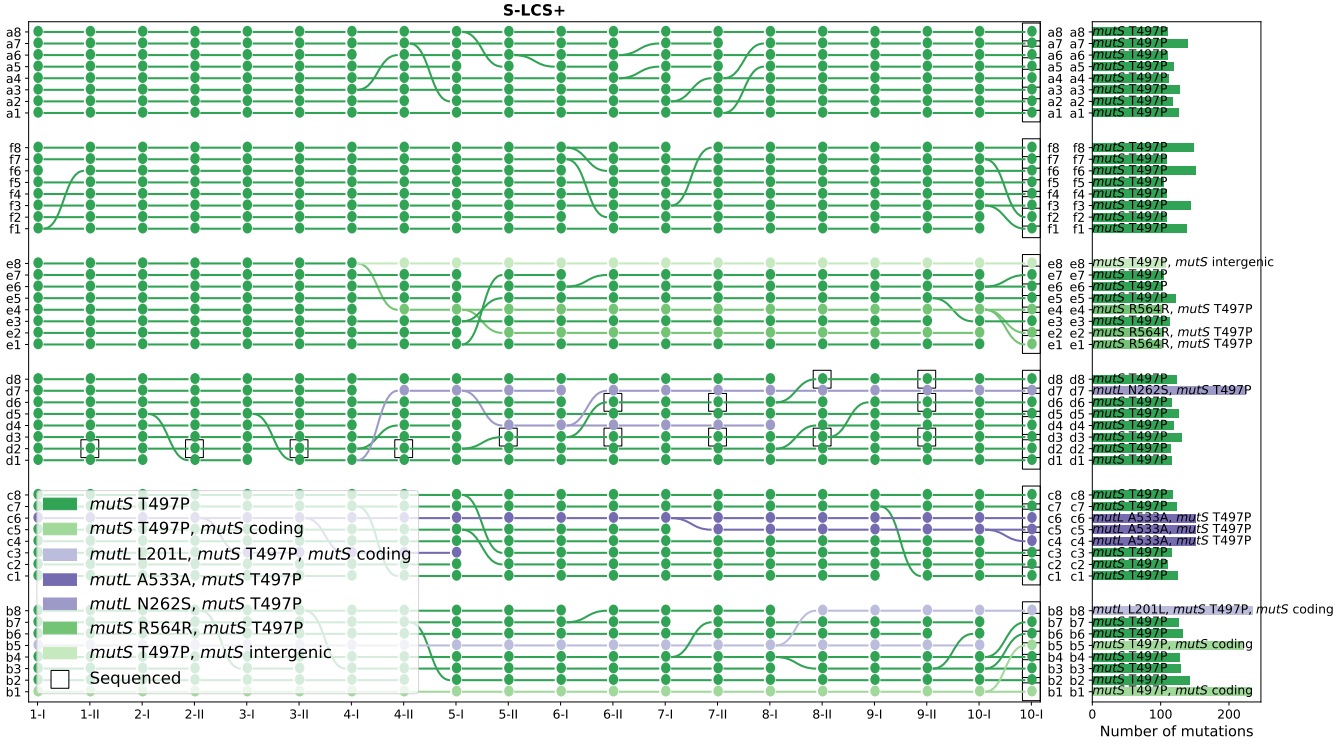


Figure S16. Mut mutations and number of mutations at the endpoint for S-LCS^{+}

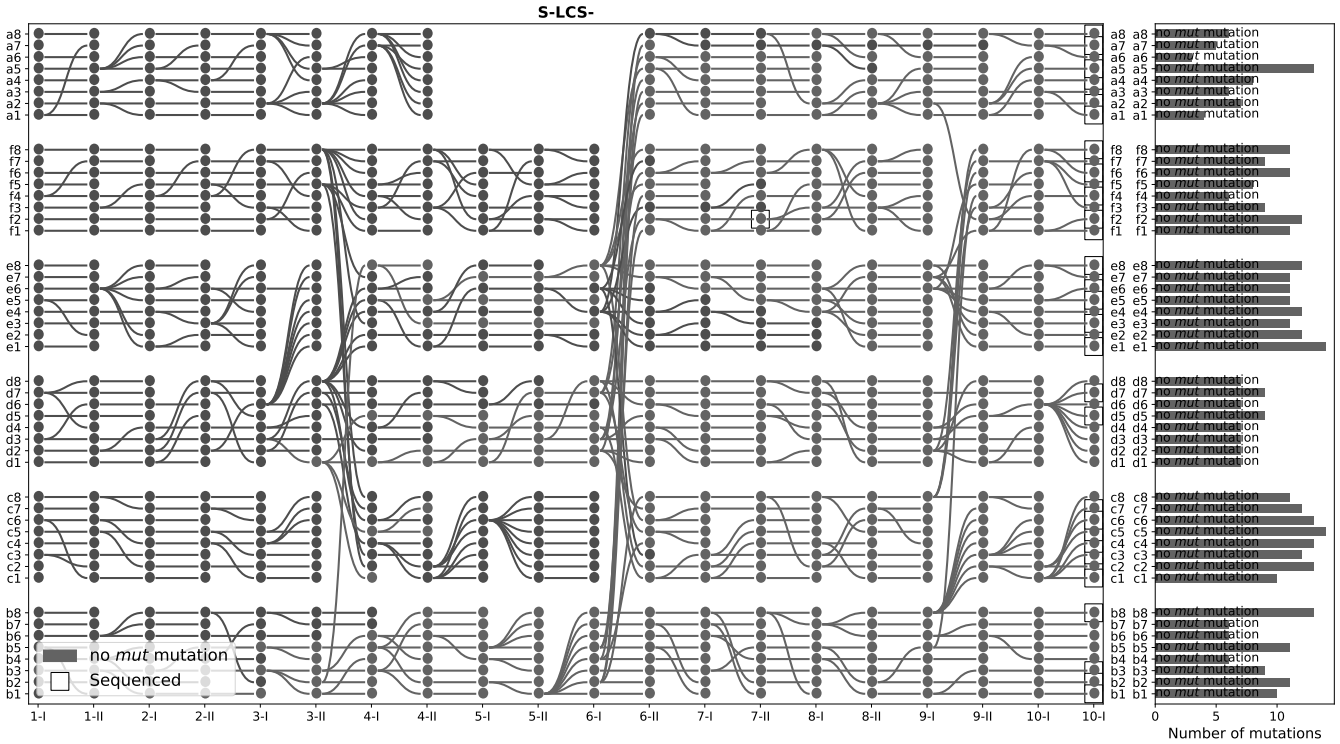


Figure S17. Mut mutations and number of mutations at the endpoint for S-LCS^{-}

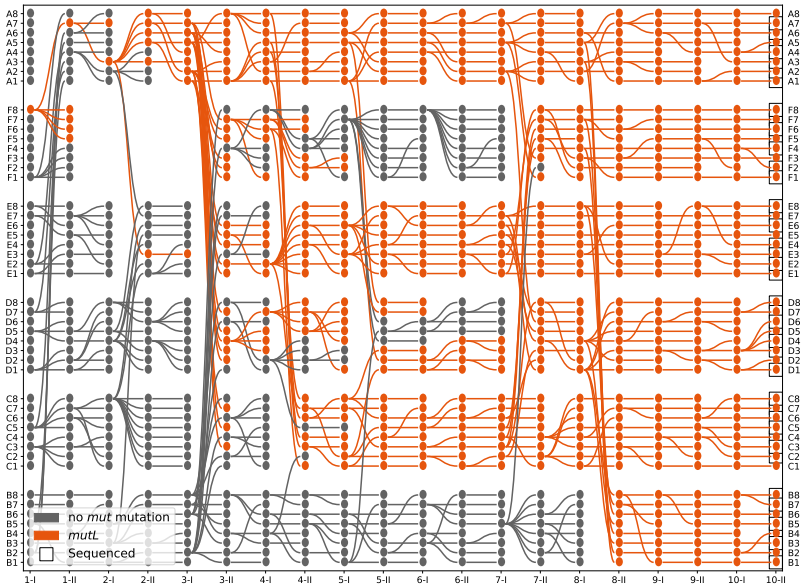


Figure S18. Origin of the *MutL* L103R mutation, estimated using only the sequencing data from the endpoints.

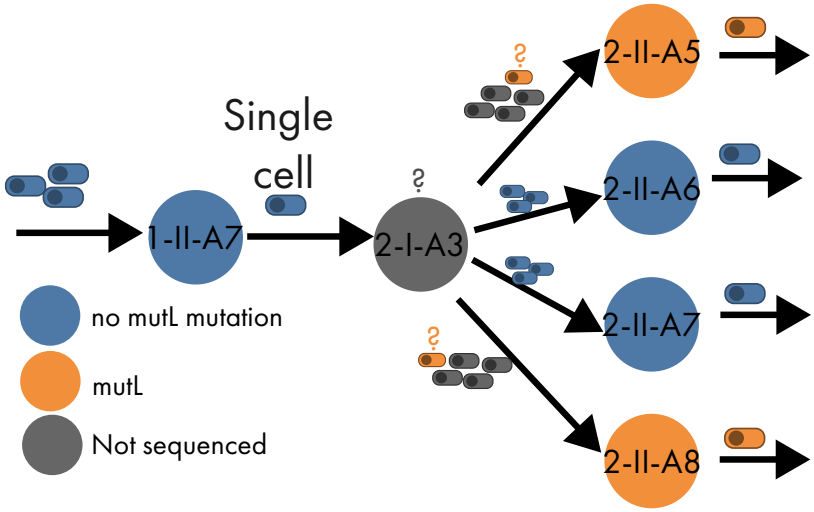


Figure S19. Sequencing result around 2-II-A5

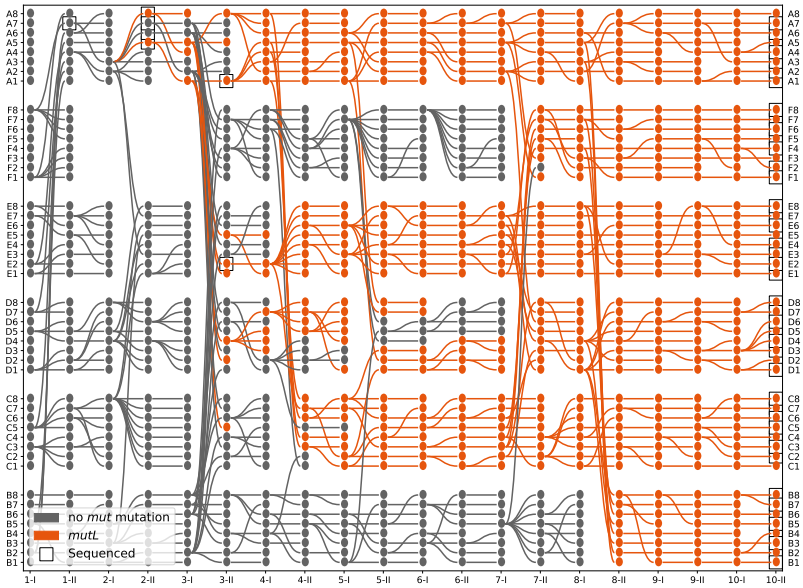


Figure S20. Origin of the *MutL* L103R mutation, estimated using the sequencing data from both the endpoints and within the genealogy.

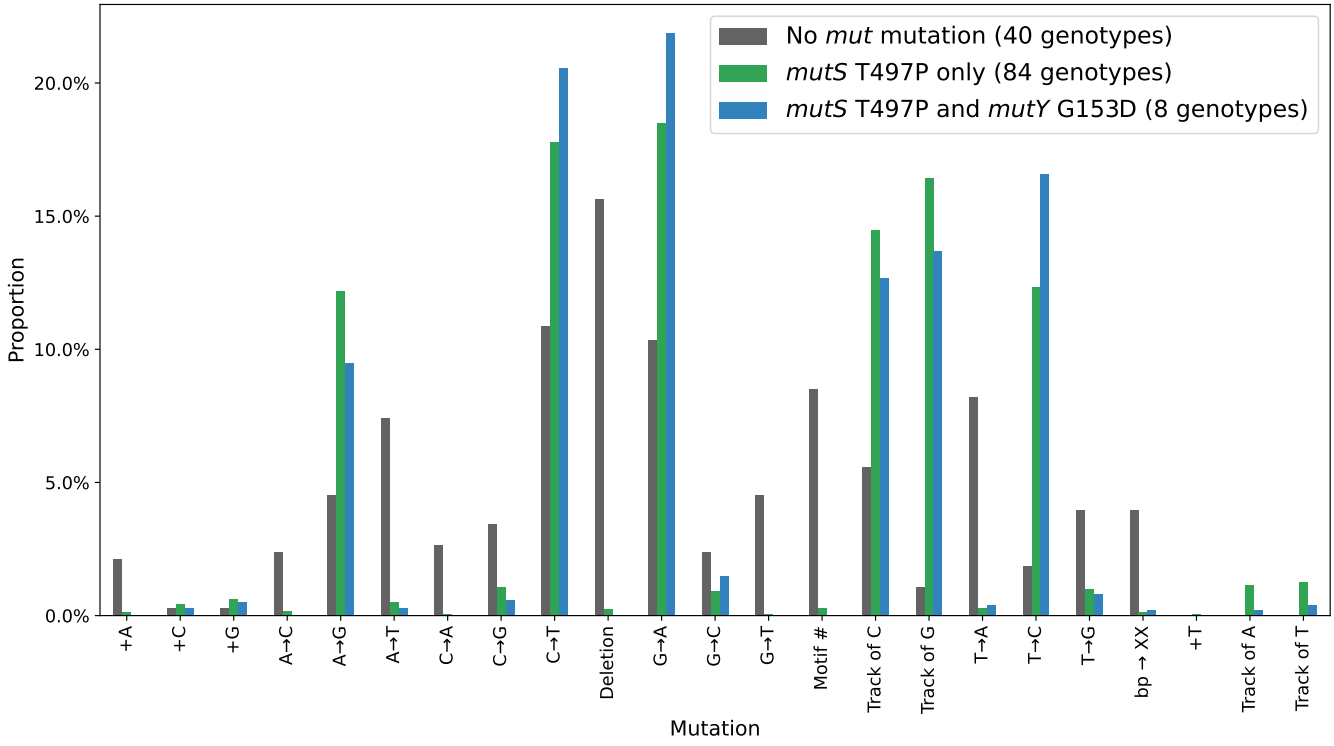


Figure S21. Kind of mutations in microcosms with no mut mutation, with only *mutS* T497P and with both *mutS* T497P and *mutY* G153D mutations.

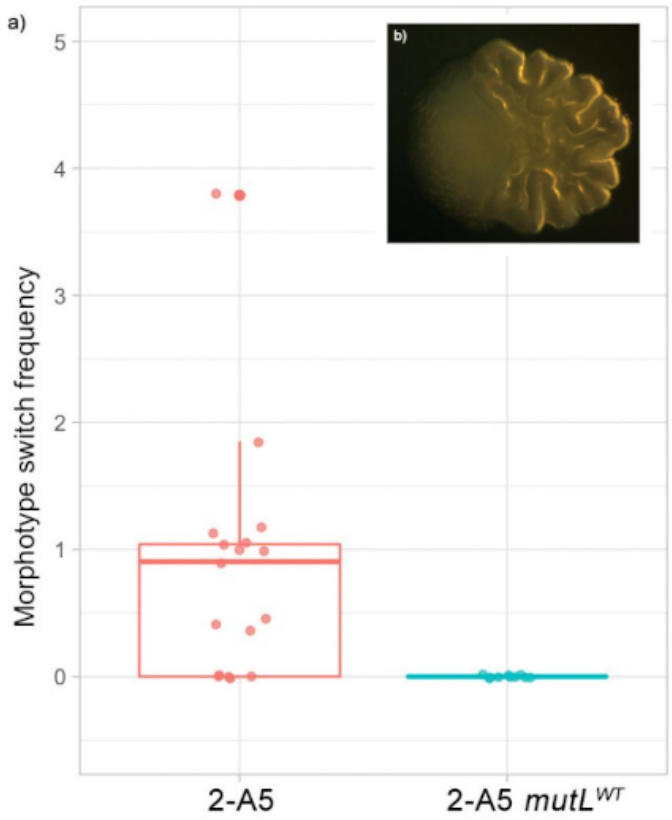


Figure S22. Colony morphology in 2-A5 Frequency of sectored colonies on plates (100-300 CFUs, 48h incubation) with switching *mutL* (L103R) morphotype in 2-A5 and 2-A5 with reconstituted *mutL* a); representative example of sectored colony showing morphotypic switch b).